LIFTING THROUGH A SMOOTH STACK OVER A FIRST-ORDER THICKENING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1990-1998 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2000-2022, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 40 of queue_stacks-stacks-more-morphisms.json (pb=1151, stmt 1990-1998, proof 2000-2022).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04B1: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2022–2022. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-local-lift-enough: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1796–1803. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04AM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2022–2022. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $T \to T'$ be a first order thickening of affine schemes. Let $\mathcal{X}'$ be an algebraic stack over T' whose structure morphism $\mathcal{X}' \to T'$ is smooth. Let $x : T \to \mathcal{X}'$ be a morphism over T'. Then there exists a morphsm $x' : T' \to \mathcal{X}'$ over T' with $x'|_T = x$.*

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We may apply the result of Lemma lemma-local-lift-enough. Thus it suffices to construct a smooth surjective morphism $W' \to T'$ with W' affine such that $x|_{T \times_{W'} T'}$ lifts to W'. (We urge the reader to find their own proof of this fact using the analogous result for algebraic spaces already established.) We choose a scheme U' and a surjective smooth morphism $U' \to \mathcal{X}'$. Observe that $U' \to T'$ is smooth and that the projection $T \times_{\mathcal{X}'} U' \to T$ is surjective smooth. Choose an affine scheme W and an étale morphism $W \to T \times_{\mathcal{X}'} U'$ such that $W \to T$ is surjective. Then $W \to T$ is a smooth morphism of affine schemes. After replacing W by a disjoint union of principal affine opens, we may assume there exists a smooth morphism of affines $W' \to T'$ such that $W = T \times_{T'} W'$, see Algebra, Lemma 04B1. By More on Morphisms of Spaces, Lemma 04AM we can find a morphism $W' \to U'$ over T' lifting the given morphism $W \to U'$. This finishes the proof. □